

Exercise - 3

a. Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method.

```
// Define a class
class Student {
    // Data members (instance variables)
    String name;
    int age;

    // Method to set student details
    void setDetails(String n, int a) {
        name = n;
        age = a;
    }

    // Method to display student details
    void displayDetails() {
        System.out.println("Student Name: " + name);
        System.out.println("Student Age: " + age);
    }
}

public class ClassMechanismExample {
    public static void main(String[] args) {
        // Creating an object of Student class
        Student s1 = new Student();

        // Calling method to set details
        s1.setDetails("Alice", 20);

        // Calling method to display details
        System.out.println("Student Information:");
        s1.displayDetails();
    }
}
```

Expected Output:

```
Student Information:
Student Name: Alice
Student Age: 20
```

b. Write a JAVA program implements method overloading.

```
public class Calculator {  
  
    // Method to add two integers  
  
    public int add(int a, int b) {  
  
        return a + b;  
  
    }  
  
    // Method to add three integers (overloaded method)  
  
    public int add(int a, int b, int c) {  
  
        return a + b + c;  
  
    }  
  
    // Method to add two doubles (overloaded method)  
  
    public double add(double a, double b) {  
  
        return a + b;  
  
    }  
  
    // Method to concatenate two strings (overloaded method)  
  
    public String add(String s1, String s2) {  
  
        return s1 + s2;  
  
    }  
  
    public static void main(String[] args) {  
  
        Calculator calc = new Calculator();  
  
        // Calling the overloaded methods  
  
        System.out.println("Sum of two integers: " + calc.add(5, 10));  
  
        System.out.println("Sum of three integers: " + calc.add(5, 10, 15));  
  
        System.out.println("Sum of two doubles: " + calc.add(2.5, 3.7));  
  
        System.out.println("Concatenation of two strings: " + calc.add("Hello", " World"));  
  
    }  
  
}
```

Expected Output:

Sum of two integers: 15
Sum of three integers: 30
Sum of two doubles: 6.2
Concatenation of two strings: Hello World

c. Write a JAVA program to implement constructor.

```
public class MyClass {  
    // Instance variables  
  
    String name;  
  
    int id;  
  
    // Default Constructor (no arguments)  
  
    public MyClass() {  
        this.name = "Default Name";  
  
        this.id = 0;  
  
        System.out.println("Default constructor called.");  
    }  
  
    // Parameterized Constructor  
  
    public MyClass(String name, int id) {  
        this.name = name;  
  
        this.id = id;  
  
        System.out.println("Parameterized constructor called for " + name);  
    }  
  
    // Method to display object details  
  
    public void displayDetails() {  
        System.out.println("Name: " + name + ", ID: " + id);  
    }  
  
    public static void main(String[] args) {  
        // Creating an object using the default constructor  
  
        MyClass obj1 = new MyClass();  
  
        obj1.displayDetails();  
    }  
}
```

```
        System.out.println(); // For better readability in output

        // Creating an object using the parameterized constructor

        MyClass obj2 = new MyClass("John Doe", 101);

        obj2.displayDetails();

    }
}
```

Expected Output:

Default constructor called.
Name: Default Name, ID: 0

Parameterized constructor called for John Doe
Name: John Doe, ID: 101

d. Write a JAVA program to implement constructor overloading.

```
public class Person
{
    // Declaring a non-parameterized constructor.
    Person() {
        System.out.println("Introduction:");
    }
    // Declaring one parameterized constructor.
    Person(String name){
        System.out.println("Name: " +name);
    }
    // Declaring two parameterized constructor.
    Person(String scname, int rollNo) {
        System.out.println("School name: "+scname+ ", "+"Roll no:"+rollNo);
    }
    public static void main(String[] args)
    {
        // JVM will call constructor depending on arguments passed.
        Person p1 = new Person(); // calling with zero argument.
        Person p2 = new Person("John"); // calling with one argument.
        Person p3 = new Person("DPS", 12); // calling with two arguments.
    }
}
```

Expected Output:

introduction:
Name: John
School name: DPS, Roll no:12